

Question 1 **2 marks**

101

Liam went for a ride on a Ferris wheel with a radius of 5 metres. Determine the distance he travelled if the Ferris wheel stopped after rotating 1260° .

Question 2 **4 marks**

102

Determine and simplify the term that contains x^{12} in the binomial expansion of $(2x^4 - 3)^7$.

Question 3 **a) 1 mark b) 2 marks**103
104

The normal healing of wounds can be modeled by an exponential function. The area of the wound decreases according to the formula,

$$A = A_0 e^{-0.35t}$$

where: A is the area of the wound in cm^2 , after t days,

A_0 is the initial area of the wound in cm^2 , and

t is the time, in days, after healing begins.

If a wound has an initial area of 25 cm^2 , determine:

a) the area of the wound after 3 days.

b) the amount of time it will take for the area of the wound to decrease to 4 cm^2 .

Question 4 **3 marks**

105

During a hockey practice, 4 players are wearing a green jersey and 5 players are wearing a red jersey. Determine the number of ways three players can be selected if at least two of the players must be wearing a red jersey.

Note: A calculator is not required for the remaining test questions.

Question 5

4 marks

106

Solve, algebraically, over the interval $[0, 2\pi]$.

$$2\sin^2 x + 5\cos x + 1 = 0$$

Question 6

1 mark

107

The function, $y = f(x)$, has a domain of $[-8, 4]$. The domain of the graph of $y = f(bx)$ is $[-4, 2]$.

State the value of b .

Question 7**1 mark**

108

Describe the difference between the graph of $f(x) = \frac{(x+3)(x-4)}{(x+3)}$ and the graph of $g(x) = x - 4$.

Question 8

1 mark

109

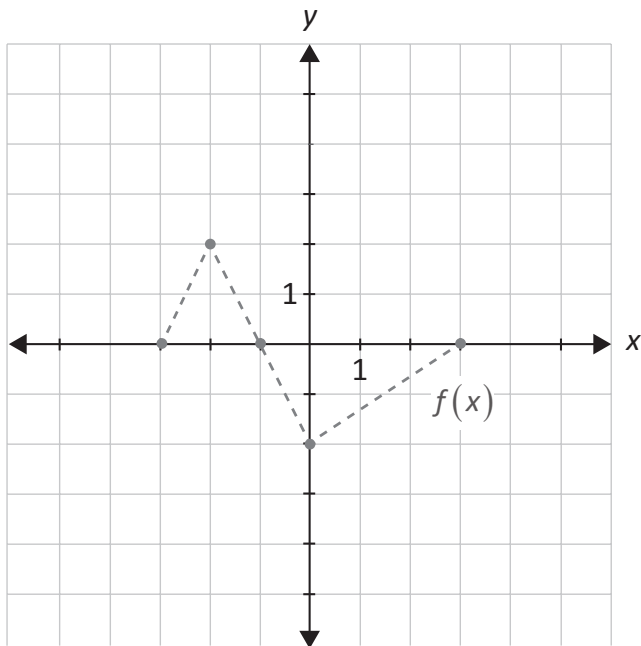
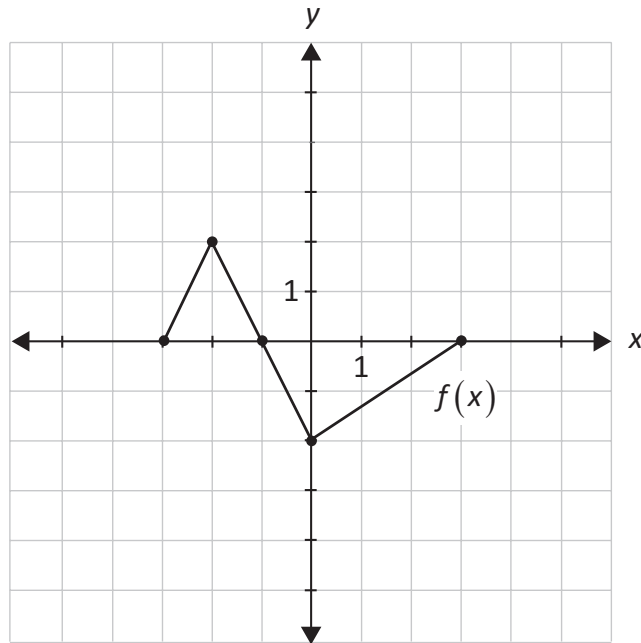
Determine if the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is on the unit circle. Justify your answer.

Question 9

3 marks

110

Given the graph of $f(x)$, sketch the graph of $g(x) = f(-(x-1)) + 3$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 10**1 mark**

111

Bonnie was asked to determine the factors of $p(x) = -3x^4 + 2x^2 - 4x + 5$.

Describe the error she made in her set up of synthetic division.

$$\begin{array}{r|rrrrr} 1 & & -3 & 2 & -4 & 5 \end{array}$$

Question 11

a) 1 mark b) 1 mark

112
113

Given the functions, $f(x) = \frac{1}{x-1}$ and $g(x) = x - 3$,

a) state the equation of $f(g(x))$.

$$f(g(x)) = \underline{\hspace{2cm}}$$

b) state the domain of $f(g(x))$.

Domain:

Question 12

3 marks

114

Prove the identity for all permissible values of x .

$$\tan 2x = \frac{2}{\cot x - \tan x}$$

Left-Hand Side	Right-Hand Side

Question 13

a) 1 mark b) 1 mark

115
116

Given $f(x) = -x^2 + 2$ and $g(x) = x^2 - 1$,

a) determine the equation of $h(x) = g(x) - f(x)$.

$h(x) =$ _____

b) state the range of $h(x)$.

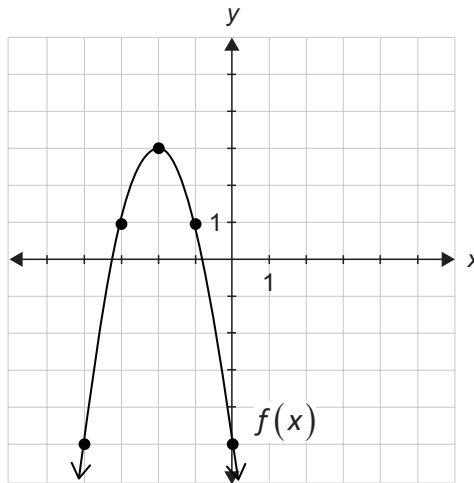
Range: _____

Question 14

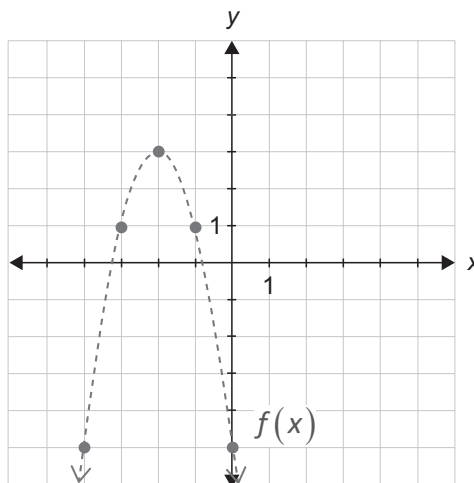
a) 1 mark b) 1 mark

117
118

Given the graph of $f(x)$,



a) sketch the graph after a reflection over the line $y = x$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

b) state a restriction on the domain of $f(x)$ in order for its inverse to be a function.

Question 15

1 mark

119

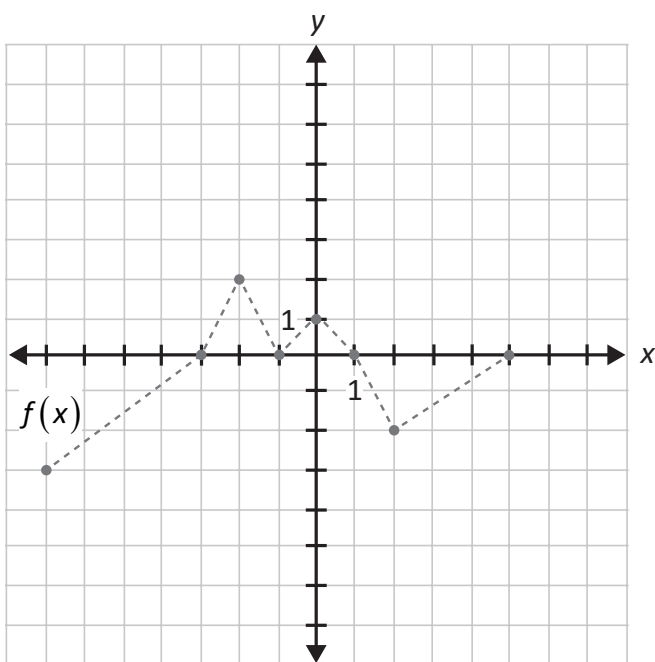
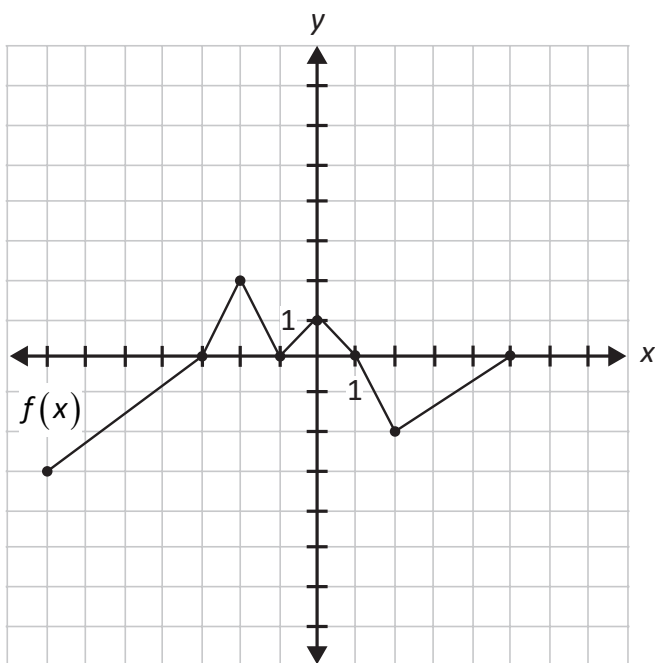
Describe a transformation of a function that would not affect its domain.

Question 16

1 mark

120

Given the graph of $f(x)$, sketch the graph of the function $y = |f(x)|$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 17

3 marks

121

Determine, algebraically, all of the zeros of the polynomial function, $p(x) = 3x^3 + x^2 - 20x + 12$.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.